

Report on
Artificial Intelligence In Cyber Security
(FullStack Development Internship)



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AI in Cyber Security

1. Introduction – AI in Cyber Security:

Cyber security means protecting computer systems, networks, and data from unauthorized access, damage, or theft. As cyberattacks are becoming more frequent and complex, older security tools are struggling to keep up. Artificial Intelligence (AI) is increasingly being used to detect, analyze, and respond to threats faster than humans alone. Studies show that most organizations now see AI as the biggest force shaping cyber security in 2026.

Traditional cyber security relied heavily on fixed rules, known virus signatures, and manual monitoring by human analysts. This approach worked well when attacks were simple and predictable, but modern attackers use automation, encryption, and constantly changing tactics that can easily slip past static defenses. AI-based security systems, on the other hand, learn continuously from data and can recognize subtle patterns that indicate an attack, even if that exact attack has never been seen before. This shift from reactive, manual defense to proactive, intelligent defense is what makes AI such an important part of modern cyber security strategy.

2. How AI is Used in Cyber Security:

- **Threat Detection** – AI studies normal user and network behaviour and flags anything unusual, such as an employee suddenly downloading large volumes of sensitive files or logging in from an unfamiliar location at odd hours.
- **Malware & Phishing Detection** – AI scans emails, links, and files to spot phishing and malicious software, analyzing sender patterns, language, and attachments to catch scams that traditional spam filters often miss.
- **Automated Incident Response** – AI can isolate affected devices or block access as soon as a threat is confirmed, containing an attack within seconds instead of the hours it might take a human team to react.
- **Predicting Future Attacks** – AI studies past incidents to estimate which systems may be targeted next, helping organizations patch weaknesses before attackers can exploit them.
- **Vulnerability Management** – AI continuously scans networks and applications to find outdated software, misconfigurations, and weak points that could be exploited, and prioritizes which issues need urgent attention.
- **User Authentication** – AI-powered biometric and behavioural authentication systems verify identity through typing patterns, facial recognition, or voice, adding an extra layer of security beyond passwords.

3. Benefits of AI in Cyber Security:

- **Speed** – Detects and responds to threats in real time, often within seconds, reducing the window of opportunity attackers have to cause damage.
- **Accuracy** – Reduces false alarms by learning normal behaviour, allowing security teams to focus on genuine threats instead of chasing false leads.
- **Scalability** – Can monitor millions of events across large networks at once, something that would be impossible for human teams to do manually.
- **Cost saving** – Reduces the workload on human security analysts, allowing organizations to allocate their skilled staff to more strategic tasks.
- **Continuous learning** – Improves over time as it sees more data and new attack types, becoming more effective the longer it is used.
- **Proactive defense** – Instead of only reacting after damage occurs, AI helps organizations anticipate and prevent attacks before they happen.

4. Challenges of AI in Cyber Security:

- **Data dependency** – AI models need large amounts of quality data to work well, and poor or biased data can lead to inaccurate detection.
- **False positives/negatives** – AI can still miss attacks or raise wrong alerts, which can cause alert fatigue among security teams.
- **AI-powered attacks** – Hackers are also using AI to create smarter malware and phishing content, creating a constant arms race between attackers and defenders.
- **High cost** – Advanced AI security tools can be expensive for small organizations, making it harder for them to keep pace with larger enterprises.
- **Lack of skilled staff** – Many organizations lack trained people to manage AI security systems, which limits how effectively these tools can be used.
- **Explainability** – AI decisions can sometimes be difficult to interpret, making it harder for analysts to understand exactly why a system flagged a particular activity as suspicious.

5. Future Scope of AI in Cyber Security:

The future scope of AI in cyber security is very promising. Security systems will become more autonomous, able to detect and fix problems with little human help. Organizations will move towards better collaboration between different AI security tools instead of using separate,

disconnected systems, and industry reports predict that most large enterprises will shift to this kind of connected, platform-based security setup within the next couple of years.

Experts also believe that as AI keeps improving, the time attackers need to find and use a software weakness will keep shrinking, so companies will have to patch and respond much faster than before. At the same time, protecting employee logins and identities is expected to become just as important as protecting networks, since AI can help attackers spot weak or reused passwords very quickly.

Governments are also expected to introduce more rules and guidelines for using AI responsibly in security, similar to early steps already being taken through regulations like the EU AI Act. New protection methods will be needed as quantum computing develops, since it may affect current encryption methods. Experts also warn that not every organization will be able to keep up with these changes at the same pace, which could create a bigger gap between well-protected companies and those that fall behind.

Looking further ahead, AI is expected to be built directly into everyday security tools rather than being treated as a separate add-on, making intelligent protection a standard part of how organizations operate. Overall, there will be growing demand for professionals trained in both AI and cyber security, as AI continues to play a bigger role in keeping systems and data safe.

6. Real-World Examples of AI in Cyber Security:

- **AI in Surveillance & Security** – News reports from 2026 describe how AI was used for large-scale surveillance and tracking during the Israel-Iran conflict. AI studied footage from public camera networks to identify movement patterns of specific individuals, and combined data from several sources (cameras, communications, satellite images) to build a clearer picture much faster than manual review. This case also raised concerns about limited human oversight when AI is used for high-stakes decisions.
- **Hackers Using AI to Build Malware** – Security reports also show hackers using AI to build malware that is faster and harder to catch. AI can create many different versions of the same malware within minutes, so each copy looks slightly different to antivirus software. Researchers have found malware that uses AI to automatically rewrite and hide its own code while running, and AI is also being used to find unknown security weaknesses (zero-day bugs) much faster than manual methods. Some hacker groups have openly said they used AI tools to help build parts of their malware. This shows why cyber security teams must also use AI-based defenses to keep up with AI-powered attacks.

- AI in Banking Fraud Detection – Many banks now use AI systems that monitor transactions in real time, comparing each transaction against a customer's typical spending habits to instantly flag and block suspicious activity, preventing fraud before money is lost.

Conclusion

Artificial Intelligence has become a vital part of modern cyber security, helping organizations detect threats faster, respond more effectively, and stay ahead of increasingly sophisticated attackers. Its ability to analyze massive amounts of data, learn from new attack patterns, and automate responses gives defenders a significant advantage that was not possible with traditional security methods.

However, AI is not a complete solution on its own. Challenges such as data dependency, high costs, lack of skilled professionals, and the rise of AI-powered attacks mean that human expertise and oversight remain essential. As AI technology continues to evolve and regulations mature, the organizations that successfully combine AI capabilities with skilled human judgment will be best positioned to protect their systems and data in the years ahead.

7. References:

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